Monotonicity for the sequent calculus

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:monotonicity of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the sequent-calculus derivability relation and its finite-premise form
- subset relations between sets of sentences

Proposition 0.1 (Monotonicity)**.** *If* $\Gamma \subseteq \Delta$ *and* $\Gamma \vdash !A$*, then* $\Delta \vdash !A$*.*

Proof. Suppose $\Gamma \vdash !A$, i.e., there is a finite $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0 \Rightarrow !A$ is !!derivable. Since $\Gamma \subseteq \Delta$, then Γ_0 is also a finite subset of Δ. The !!derivation of $\Gamma_0 \Rightarrow !A$ thus also shows $\Delta \vdash !A$. □

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